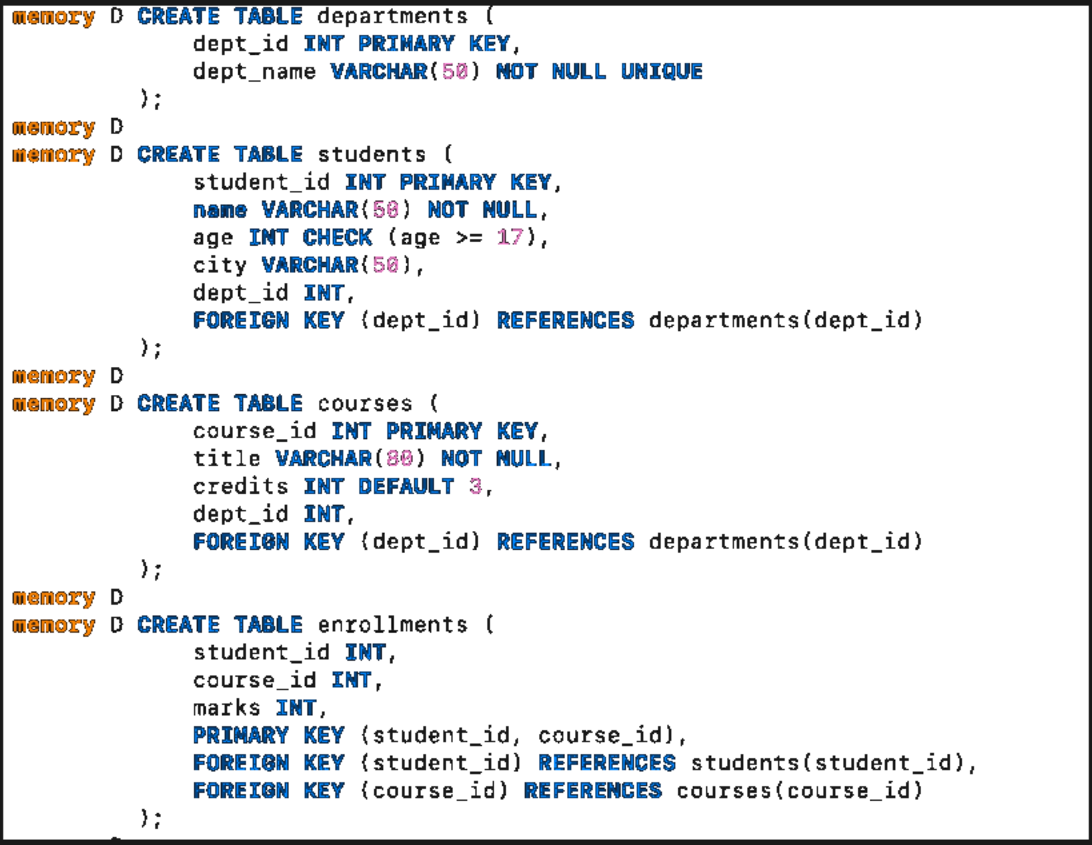
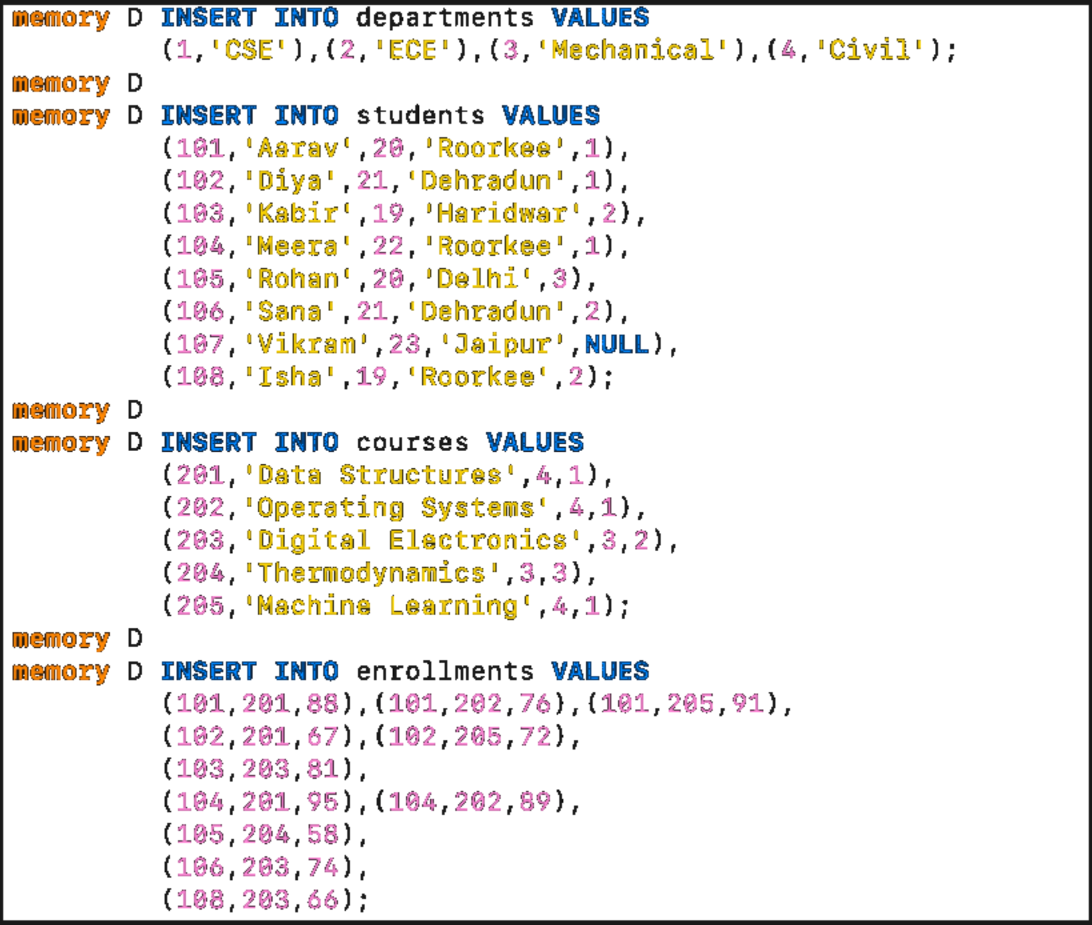
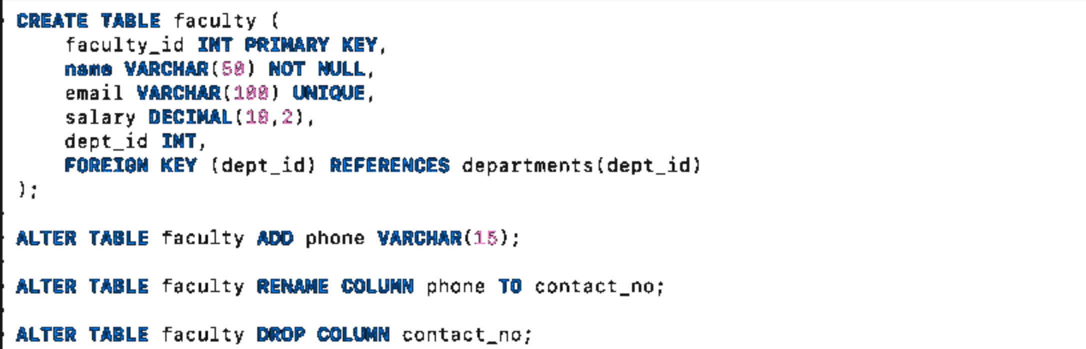


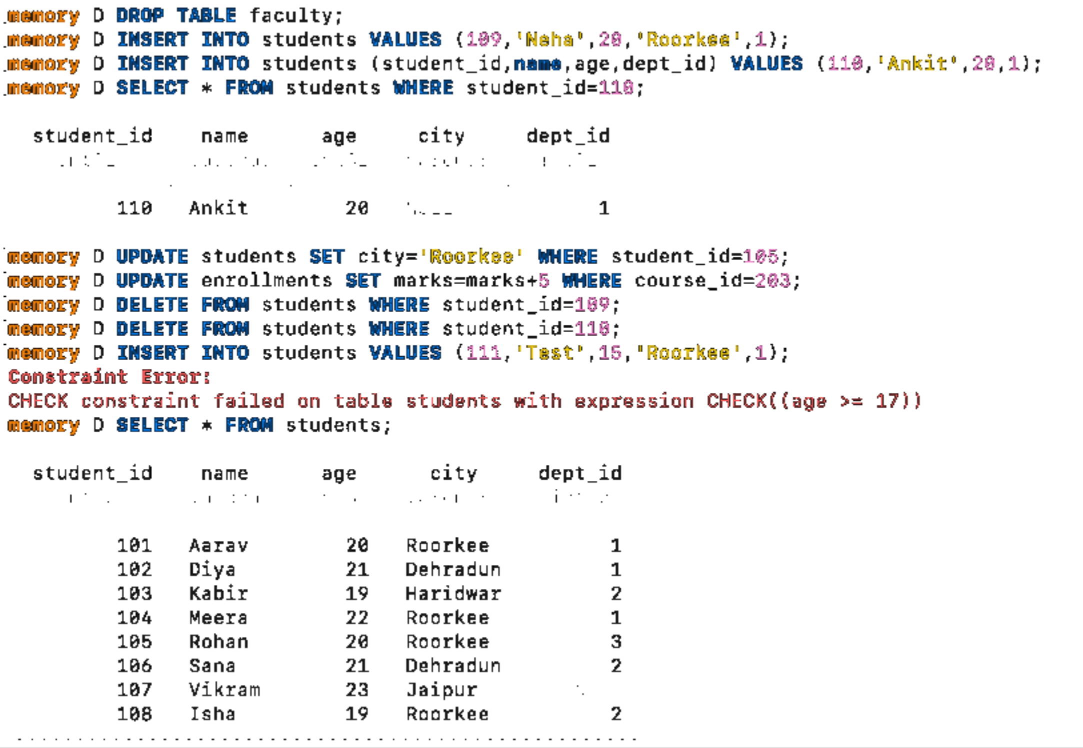
```
memory D CREATE TABLE departments (
             dept_id INT PRIMARY KEY,
             dept_name VARCHAR(50) NOT NULL UNIQUE
         );
memory D
memory D CREATE TABLE students (
             student_id INT PRIMARY KEY,
             name VARCHAR(50) NOT NULL,
             age INT CHECK (age >= 17),
             city VARCHAR(50),
             dept_id INT,
             FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
         );
memory D
memory D CREATE TABLE courses (
             course_id INT PRIMARY KEY,
             title VARCHAR(80) NOT NULL,
             credits INT DEFAULT 3,
             dept_id INT,
             FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
         );
memory D
memory D CREATE TABLE enrollments (
             student_id INT,
             course_id INT,
             marks INT,
             PRIMARY KEY (student_id, course_id),
             FOREIGN KEY (student_id) REFERENCES students(student_id),
             FOREIGN KEY (course_id) REFERENCES courses(course_id)
         );
```



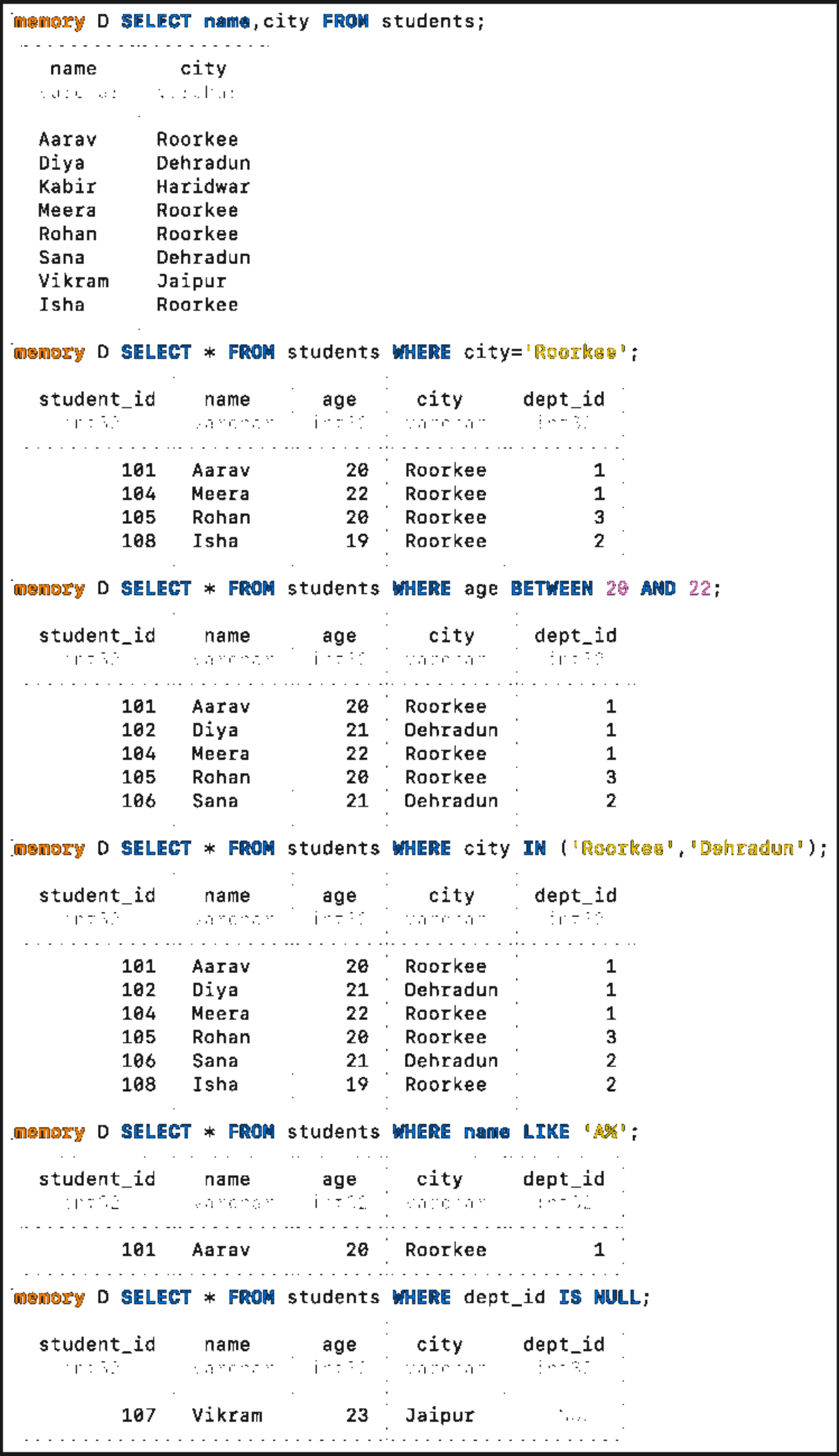
```
memory D INSERT INTO departments VALUES
         (1,'CSE'),(2,'ECE'),(3,'Mechanical'),(4,'Civil');
memory D
memory D INSERT INTO students VALUES
         (101,'Aarav',20,'Roorkee',1),
         (102,'Diya',21,'Dehradun',1),
         (103,'Kabir',19,'Haridwar',2),
         (104,'Meera',22,'Roorkee',1),
         (105,'Rohan',20,'Delhi',3),
         (106,'Sana',21,'Dehradun',2),
         (107,'Vikram',23,'Jaipur',NULL),
         (108,'Isha',19,'Roorkee',2);
memory D
memory D INSERT INTO courses VALUES
         (201,'Data Structures',4,1),
         (202,'Operating Systems',4,1),
         (203,'Digital Electronics',3,2),
         (204,'Thermodynamics',3,3),
         (205,'Machine Learning',4,1);
memory D
memory D INSERT INTO enrollments VALUES
         (101,201,88),(101,202,76),(101,205,91),
         (102,201,67),(102,205,72),
         (103,203,81),
         (104,201,95),(104,202,89),
         (105,204,58),
         (106,203,74),
         (108,203,66);
```



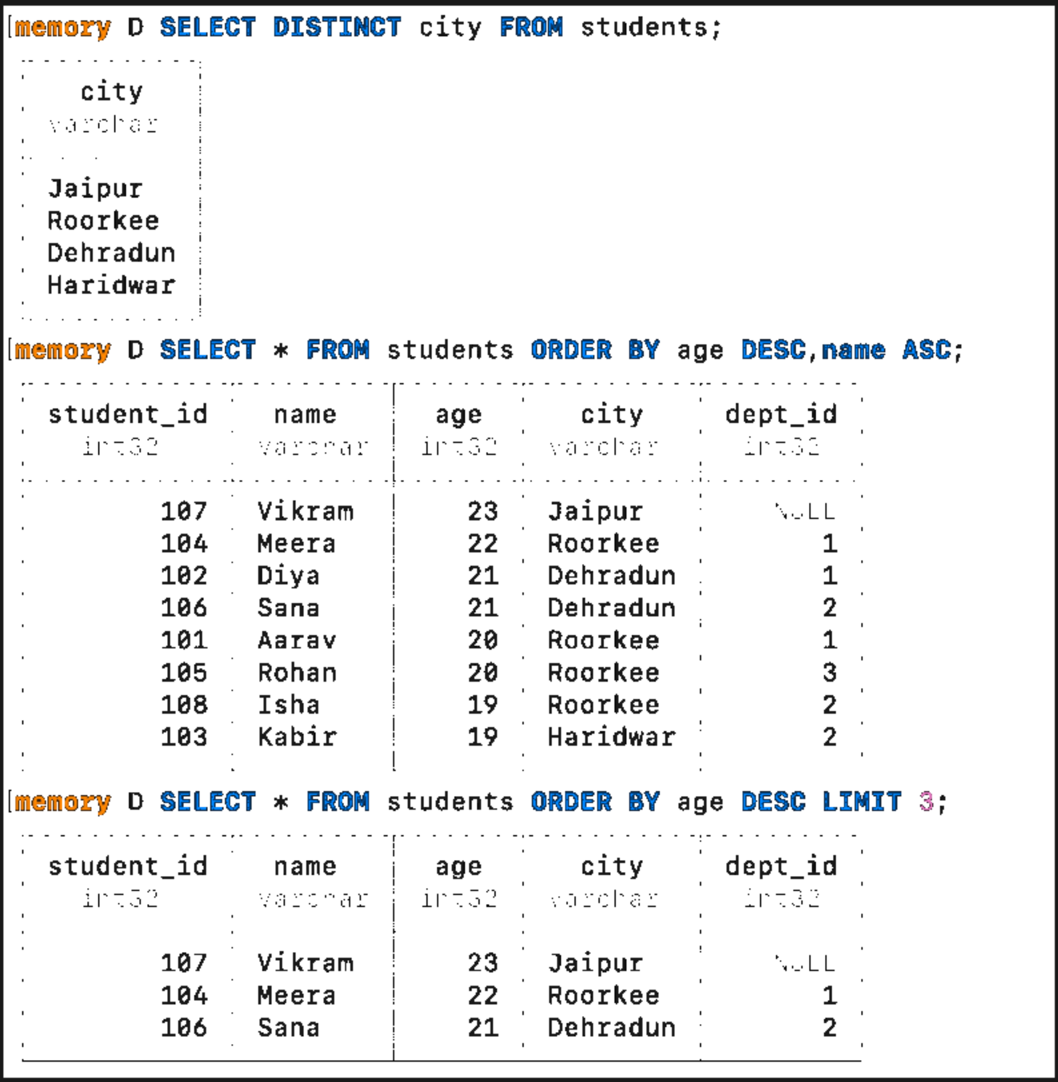
```
CREATE TABLE faculty (
    faculty_id INT PRIMARY KEY,
    name VARCHAR(50) NOT NULL,
    email VARCHAR(100) UNIQUE,
    salary DECIMAL(10,2),
    dept_id INT,
    FOREIGN KEY (dept_id) REFERENCES departments(dept_id)
);

ALTER TABLE faculty ADD phone VARCHAR(15);

ALTER TABLE faculty RENAME COLUMN phone TO contact_no;

ALTER TABLE faculty DROP COLUMN contact_no;
DROP removes table structure and data; TRUNCATE removes all rows but keeps structure;
```



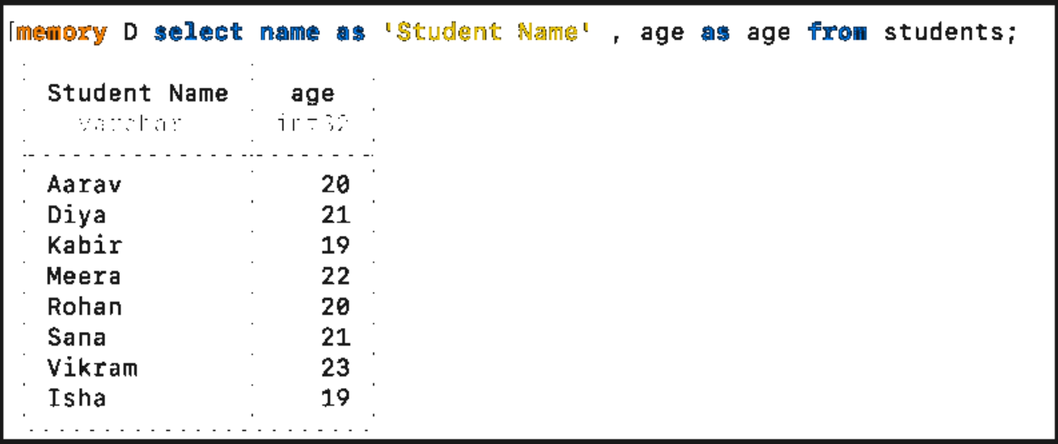
```
memory D DROP TABLE faculty;
memory D INSERT INTO students VALUES (109,'Neha',20,'Roorkee',1);
memory D INSERT INTO students (student_id,name,age,dept_id) VALUES (110,'Ankit',20,1);
memory D SELECT * FROM students WHERE student_id=110;

  student_id    name       age     city      dept_id

         110   Ankit        20   [illegible]          1

memory D UPDATE students SET city='Roorkee' WHERE student_id=105;
memory D UPDATE enrollments SET marks=marks+5 WHERE course_id=203;
memory D DELETE FROM students WHERE student_id=109;
memory D DELETE FROM students WHERE student_id=110;
memory D INSERT INTO students VALUES (111,'Test',15,'Roorkee',1);
Constraint Error:
CHECK constraint failed on table students with expression CHECK((age >= 17))
memory D SELECT * FROM students;

  student_id    name       age      city      dept_id

         101   Aarav        20   Roorkee            1
         102   Diya         21   Dehradun           1
         103   Kabir        19   Haridwar           2
         104   Meera        22   Roorkee            1
         105   Rohan        20   Roorkee            3
         106   Sana         21   Dehradun           2
         107   Vikram       23   Jaipur
         108   Isha         19   Roorkee            2
```



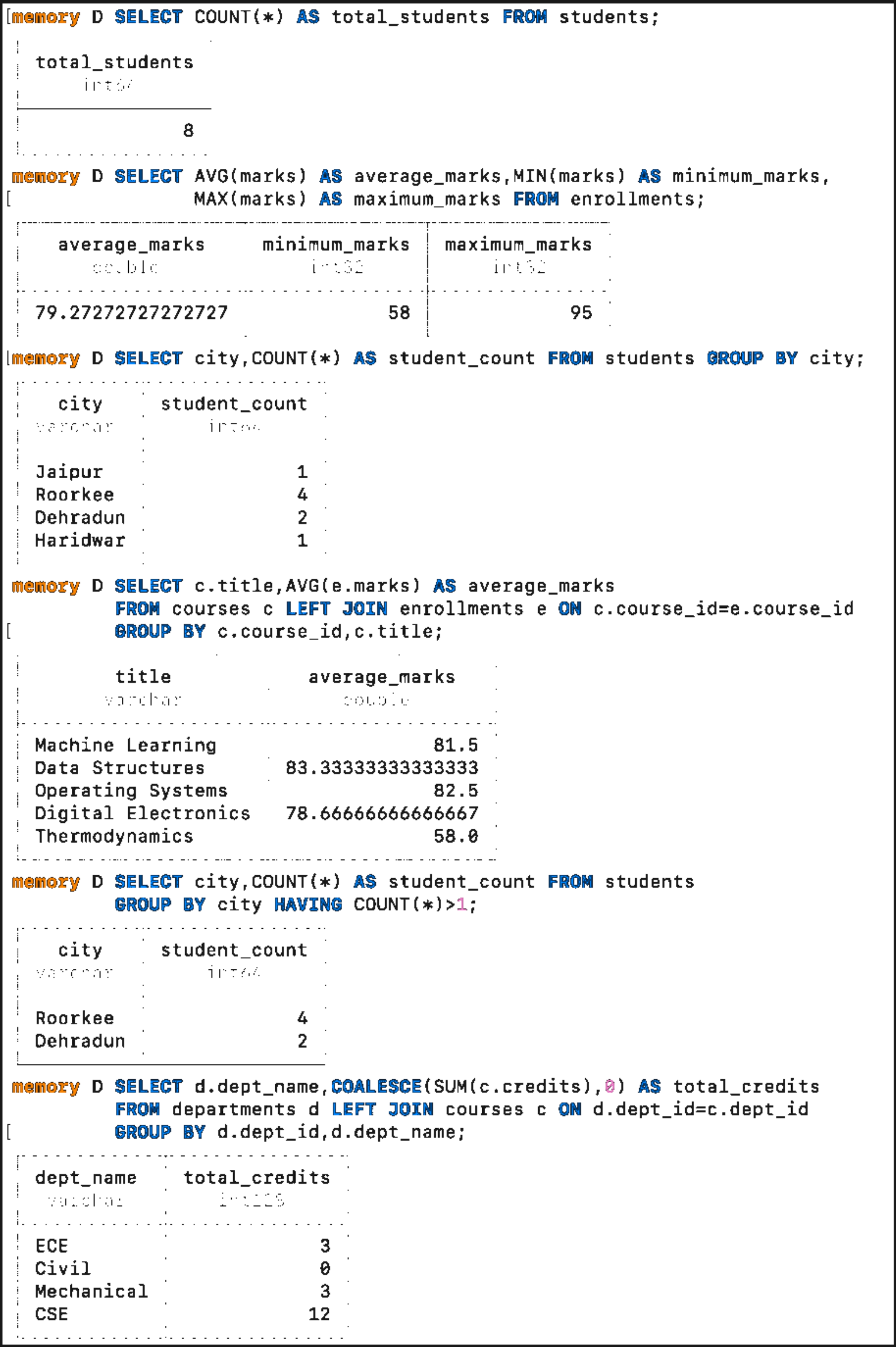
```
memory D SELECT name,city FROM students;

   name       city

  Aarav      Roorkee
  Diya       Dehradun
  Kabir      Haridwar
  Meera      Roorkee
  Rohan      Roorkee
  Sana       Dehradun
  Vikram     Jaipur
  Isha       Roorkee

memory D SELECT * FROM students WHERE city='Roorkee';

  student_id    name      age     city      dept_id


         101   Aarav       20   Roorkee           1
         104   Meera       22   Roorkee           1
         105   Rohan       20   Roorkee           3
         108   Isha        19   Roorkee           2

memory D SELECT * FROM students WHERE age BETWEEN 20 AND 22;

  student_id    name      age     city       dept_id


         101   Aarav       20   Roorkee            1
         102   Diya        21   Dehradun           1
         104   Meera       22   Roorkee            1
         105   Rohan       20   Roorkee            3
         106   Sana        21   Dehradun           2

memory D SELECT * FROM students WHERE city IN ('Roorkee','Dehradun');

  student_id    name      age     city       dept_id


         101   Aarav       20   Roorkee            1
         102   Diya        21   Dehradun           1
         104   Meera       22   Roorkee            1
         105   Rohan       20   Roorkee            3
         106   Sana        21   Dehradun           2
         108   Isha        19   Roorkee            2

memory D SELECT * FROM students WHERE name LIKE 'A%';

  student_id    name      age     city      dept_id


         101   Aarav       20   Roorkee           1

memory D SELECT * FROM students WHERE dept_id IS NULL;

  student_id    name      age     city      dept_id


         107   Vikram      23   Jaipur
```



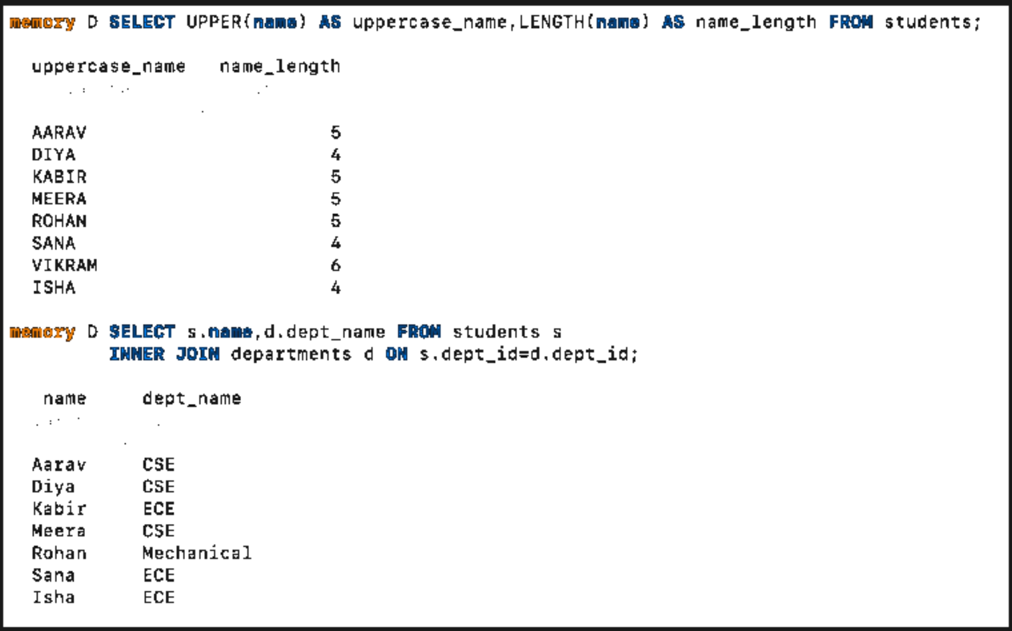
```
memory D SELECT DISTINCT city FROM students;

   city
  varchar

 Jaipur
 Roorkee
 Dehradun
 Haridwar

memory D SELECT * FROM students ORDER BY age DESC,name ASC;

 student_id    name     age    city     dept_id
   int32     varchar   int32  varchar    int32

        107  Vikram       23  Jaipur        NULL
        104  Meera        22  Roorkee          1
        102  Diya         21  Dehradun         1
        106  Sana         21  Dehradun         2
        101  Aarav        20  Roorkee          1
        105  Rohan        20  Roorkee          3
        108  Isha         19  Roorkee          2
        103  Kabir        19  Haridwar         2

memory D SELECT * FROM students ORDER BY age DESC LIMIT 3;

 student_id    name     age    city     dept_id
   int32     varchar   int32  varchar    int32

        107  Vikram       23  Jaipur        NULL
        104  Meera        22  Roorkee          1
        106  Sana         21  Dehradun         2
```



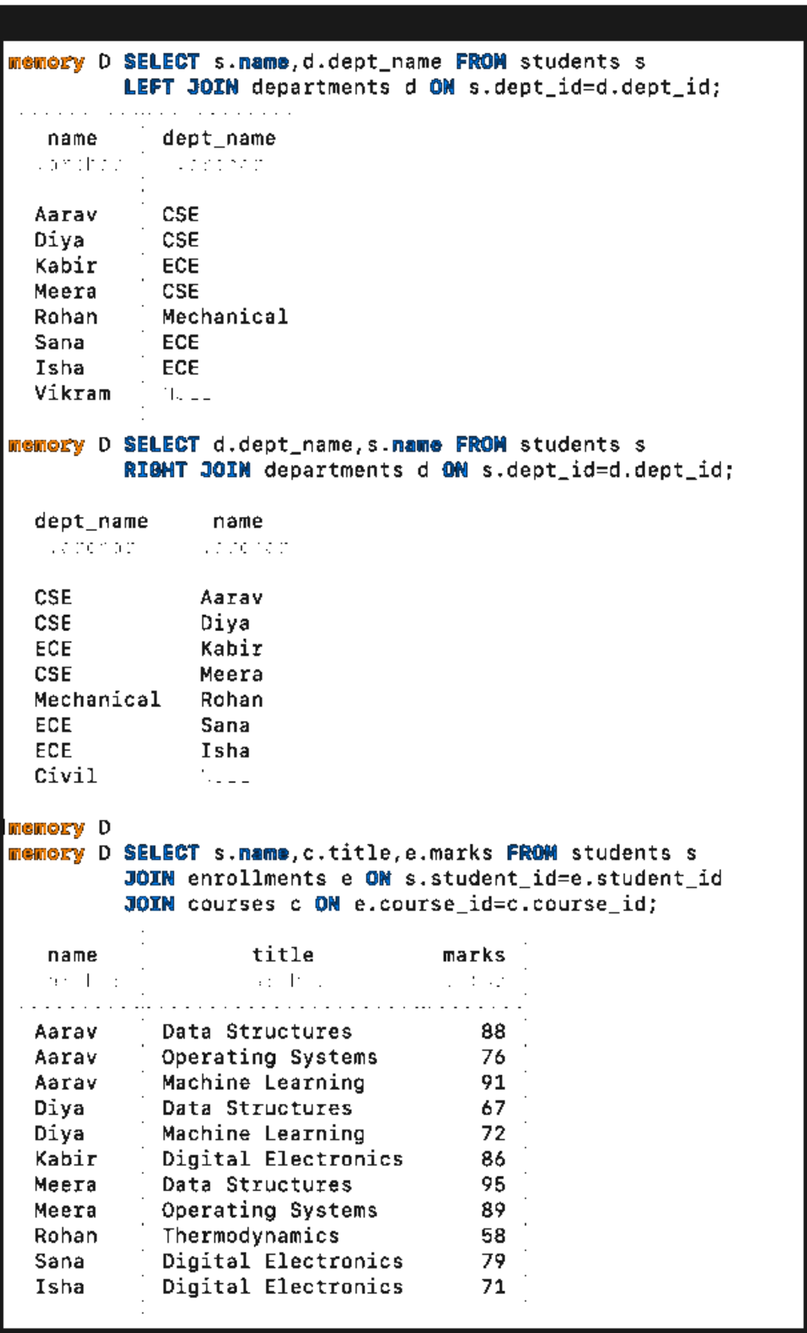
```
memory D select name as 'Student Name' , age as age from students;

 Student Name    age
   varchar      int32

 Aarav             20
 Diya              21
 Kabir             19
 Meera             22
 Rohan             20
 Sana              21
 Vikram            23
 Isha              19
```



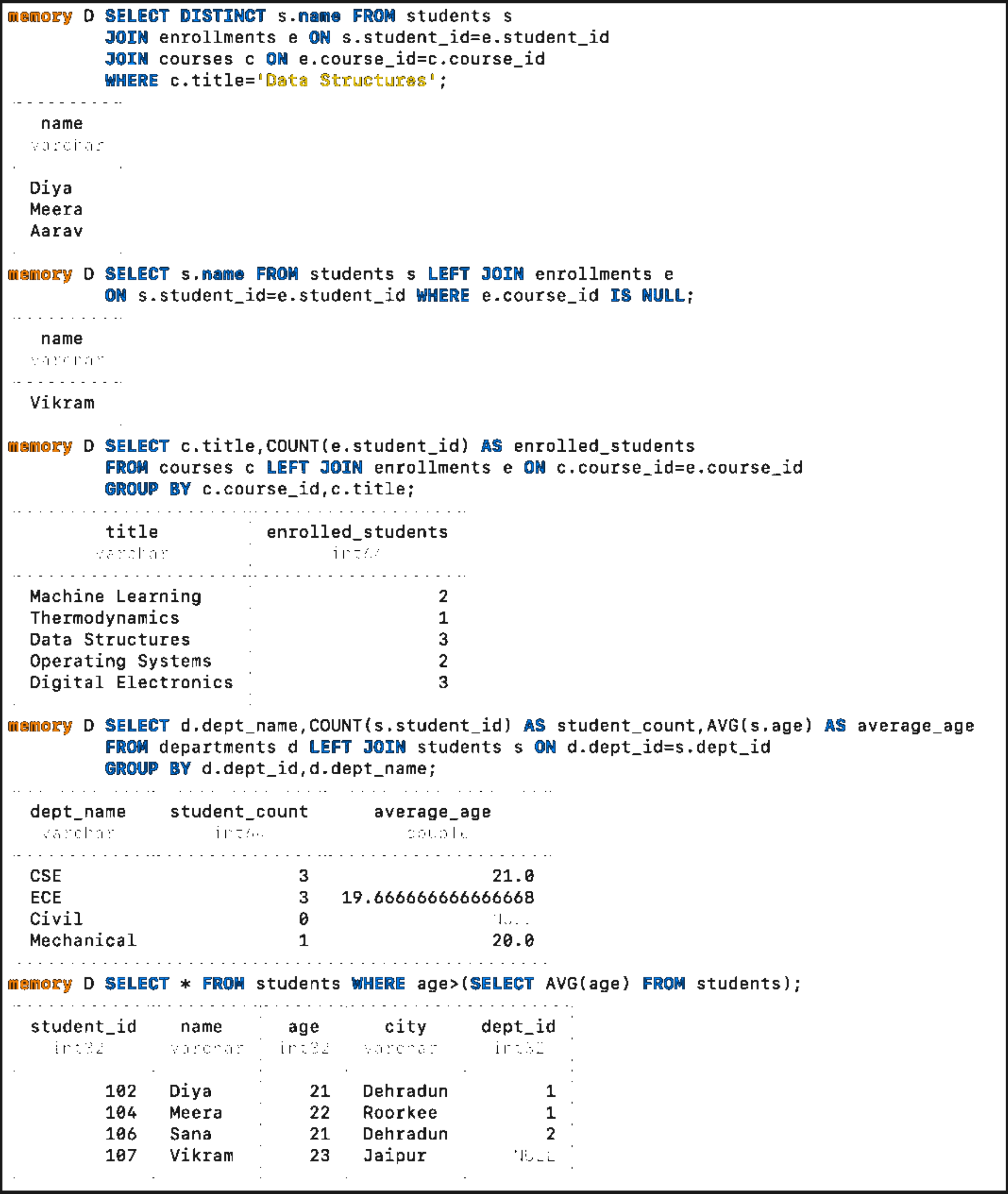
```
memory D SELECT COUNT(*) AS total_students FROM students;

  total_students
      int64

               8

memory D SELECT AVG(marks) AS average_marks,MIN(marks) AS minimum_marks,
                MAX(marks) AS maximum_marks FROM enrollments;

    average_marks     minimum_marks   maximum_marks
       double             int32           int32

  79.27272727272727              58              95

memory D SELECT city,COUNT(*) AS student_count FROM students GROUP BY city;

    city     student_count
   varchar       int64

  Jaipur                 1
  Roorkee                4
  Dehradun               2
  Haridwar               1

memory D SELECT c.title,AVG(e.marks) AS average_marks
         FROM courses c LEFT JOIN enrollments e ON c.course_id=e.course_id
         GROUP BY c.course_id,c.title;

          title              average_marks
         varchar                double

  Machine Learning                     81.5
  Data Structures        83.33333333333333
  Operating Systems                    82.5
  Digital Electronics    78.66666666666667
  Thermodynamics                       58.0

memory D SELECT city,COUNT(*) AS student_count FROM students
         GROUP BY city HAVING COUNT(*)>1;

    city     student_count
   varchar       int64

  Roorkee                4
  Dehradun               2

memory D SELECT d.dept_name,COALESCE(SUM(c.credits),0) AS total_credits
         FROM departments d LEFT JOIN courses c ON d.dept_id=c.dept_id
         GROUP BY d.dept_id,d.dept_name;

  dept_name    total_credits
   varchar        int128

  ECE                      3
  Civil                    0
  Mechanical               3
  CSE                     12
```



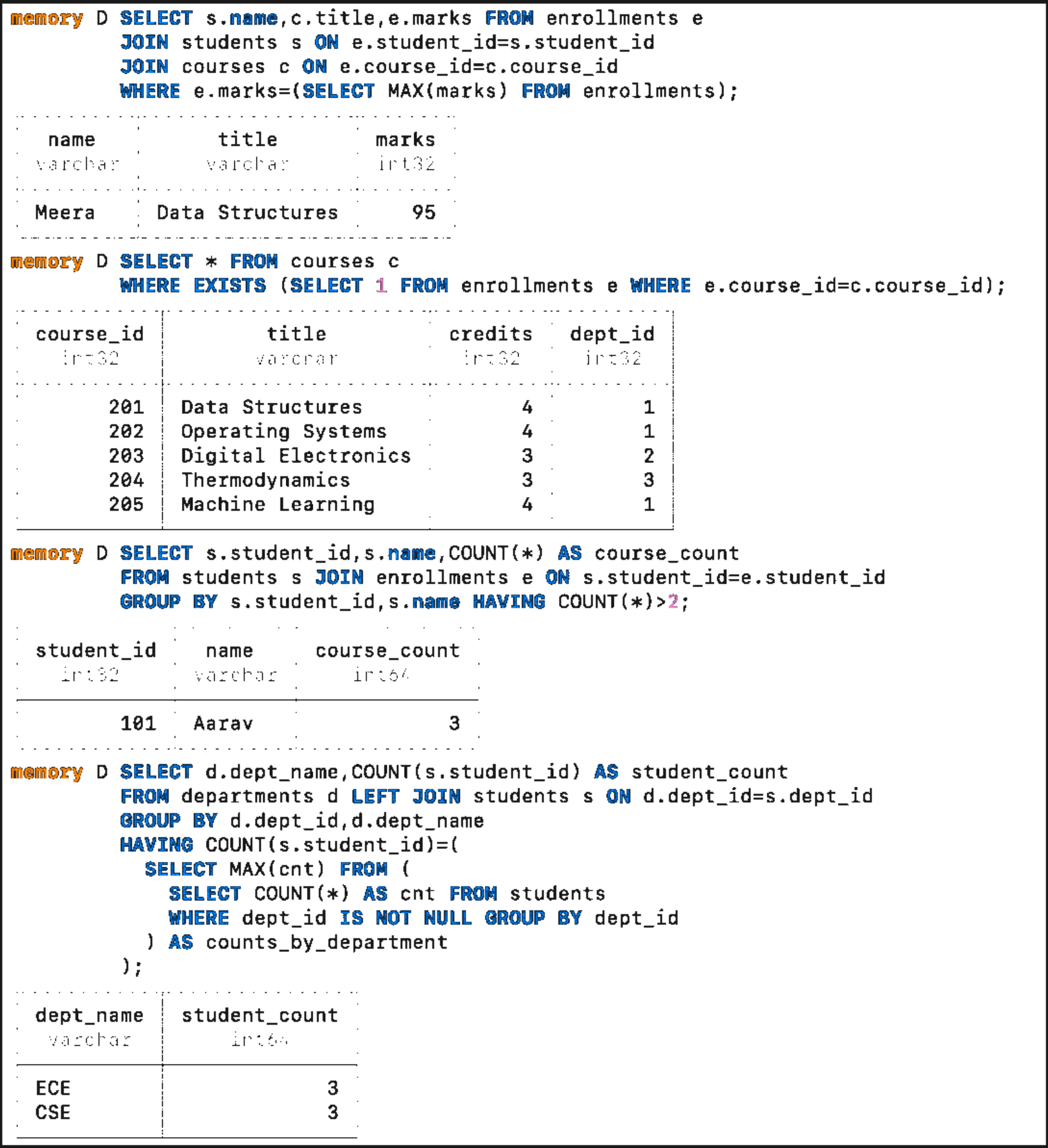
```
memory D SELECT UPPER(name) AS uppercase_name,LENGTH(name) AS name_length FROM students;

  uppercase_name   name_length


  AARAV                      5
  DIYA                       4
  KABIR                      5
  MEERA                      5
  ROHAN                      5
  SANA                       4
  VIKRAM                     6
  ISHA                       4

memory D SELECT s.name,d.dept_name FROM students s
         INNER JOIN departments d ON s.dept_id=d.dept_id;

   name     dept_name


  Aarav     CSE
  Diya      CSE
  Kabir     ECE
  Meera     CSE
  Rohan     Mechanical
  Sana      ECE
  Isha      ECE
```



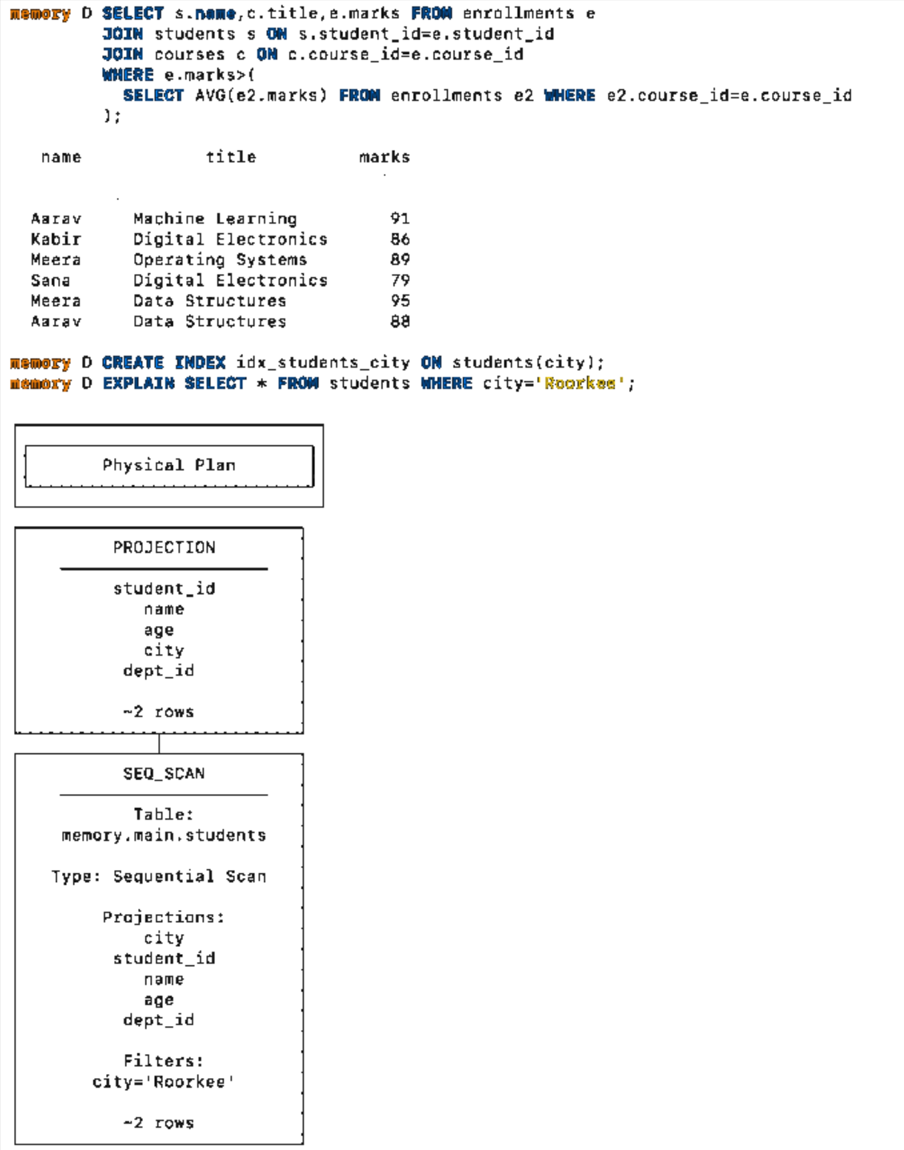
```
memory D SELECT s.name,d.dept_name FROM students s
         LEFT JOIN departments d ON s.dept_id=d.dept_id;

   name     dept_name


  Aarav     CSE
  Diya      CSE
  Kabir     ECE
  Meera     CSE
  Rohan     Mechanical
  Sana      ECE
  Isha      ECE
  Vikram    NULL

memory D SELECT d.dept_name,s.name FROM students s
         RIGHT JOIN departments d ON s.dept_id=d.dept_id;

  dept_name     name


  CSE          Aarav
  CSE          Diya
  ECE          Kabir
  CSE          Meera
  Mechanical   Rohan
  ECE          Sana
  ECE          Isha
  Civil        NULL

memory D
memory D SELECT s.name,c.title,e.marks FROM students s
         JOIN enrollments e ON s.student_id=e.student_id
         JOIN courses c ON e.course_id=c.course_id;

   name             title            marks


  Aarav     Data Structures            88
  Aarav     Operating Systems          76
  Aarav     Machine Learning           91
  Diya      Data Structures            67
  Diya      Machine Learning           72
  Kabir     Digital Electronics        86
  Meera     Data Structures            95
  Meera     Operating Systems          89
  Rohan     Thermodynamics             58
  Sana      Digital Electronics        79
  Isha      Digital Electronics        71
```



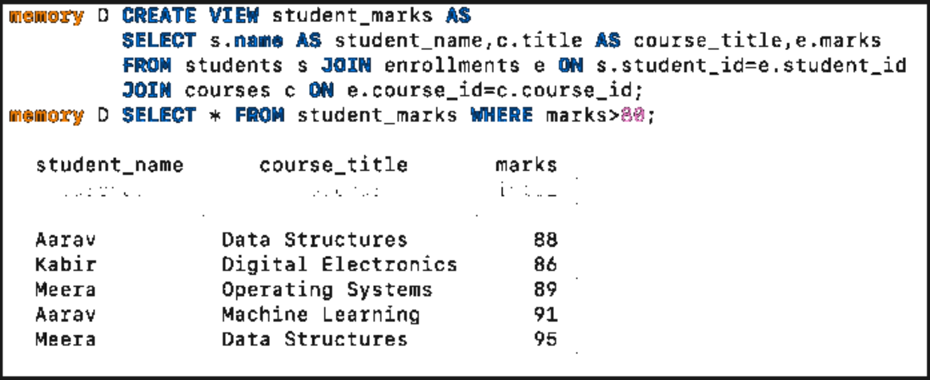
```
memory D SELECT DISTINCT s.name FROM students s
         JOIN enrollments e ON s.student_id=e.student_id
         JOIN courses c ON e.course_id=c.course_id
         WHERE c.title='Data Structures';

   name
  varchar

  Diya
  Meera
  Aarav

memory D SELECT s.name FROM students s LEFT JOIN enrollments e
         ON s.student_id=e.student_id WHERE e.course_id IS NULL;

   name
  varchar

  Vikram

memory D SELECT c.title,COUNT(e.student_id) AS enrolled_students
         FROM courses c LEFT JOIN enrollments e ON c.course_id=e.course_id
         GROUP BY c.course_id,c.title;

        title             enrolled_students
       varchar                  int64

  Machine Learning                        2
  Thermodynamics                          1
  Data Structures                         3
  Operating Systems                       2
  Digital Electronics                     3

memory D SELECT d.dept_name,COUNT(s.student_id) AS student_count,AVG(s.age) AS average_age
         FROM departments d LEFT JOIN students s ON d.dept_id=s.dept_id
         GROUP BY d.dept_id,d.dept_name;

  dept_name    student_count       average_age
   varchar         int64              double

  CSE                      3                 21.0
  ECE                      3   19.666666666666668
  Civil                    0                 NULL
  Mechanical               1                 20.0

memory D SELECT * FROM students WHERE age>(SELECT AVG(age) FROM students);

  student_id    name      age     city      dept_id
    int32     varchar    int32   varchar     int32

         102   Diya        21   Dehradun          1
         104   Meera       22   Roorkee           1
         106   Sana        21   Dehradun          2
         107   Vikram      23   Jaipur         NULL
```



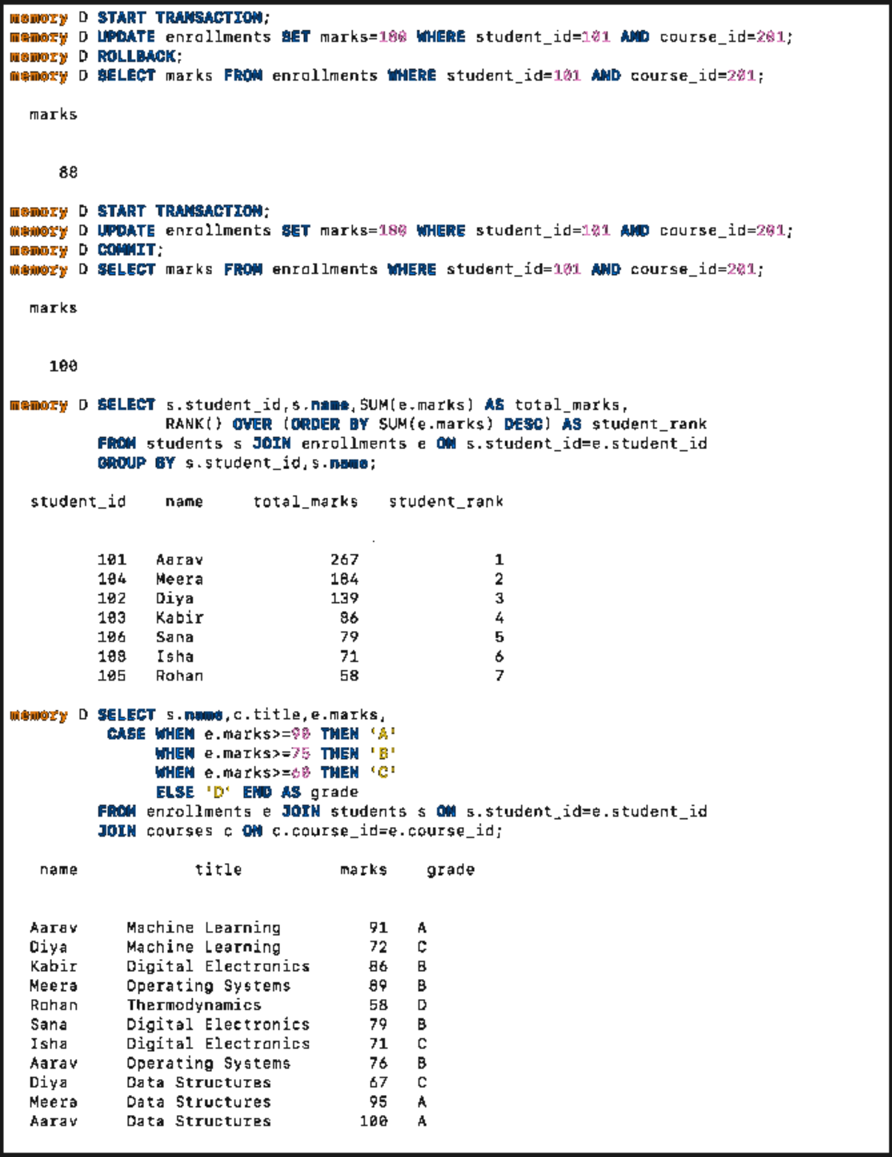
```
memory D SELECT s.name,c.title,e.marks FROM enrollments e
         JOIN students s ON e.student_id=s.student_id
         JOIN courses c ON e.course_id=c.course_id
         WHERE e.marks=(SELECT MAX(marks) FROM enrollments);
┌─────────┬─────────────────┬───────┐
│  name   │      title      │ marks │
│ varchar │     varchar     │ int32 │
├─────────┼─────────────────┼───────┤
│ Meera   │ Data Structures │    95 │
└─────────┴─────────────────┴───────┘
memory D SELECT * FROM courses c
         WHERE EXISTS (SELECT 1 FROM enrollments e WHERE e.course_id=c.course_id);
┌───────────┬─────────────────────┬─────────┬─────────┐
│ course_id │        title        │ credits │ dept_id │
│   int32   │       varchar       │  int32  │  int32  │
├───────────┼─────────────────────┼─────────┼─────────┤
│       201 │ Data Structures     │       4 │       1 │
│       202 │ Operating Systems   │       4 │       1 │
│       203 │ Digital Electronics │       3 │       2 │
│       204 │ Thermodynamics      │       3 │       3 │
│       205 │ Machine Learning    │       4 │       1 │
└───────────┴─────────────────────┴─────────┴─────────┘
memory D SELECT s.student_id,s.name,COUNT(*) AS course_count
         FROM students s JOIN enrollments e ON s.student_id=e.student_id
         GROUP BY s.student_id,s.name HAVING COUNT(*)>2;
┌────────────┬─────────┬──────────────┐
│ student_id │  name   │ course_count │
│   int32    │ varchar │    int64     │
├────────────┼─────────┼──────────────┤
│        101 │ Aarav   │            3 │
└────────────┴─────────┴──────────────┘
memory D SELECT d.dept_name,COUNT(s.student_id) AS student_count
         FROM departments d LEFT JOIN students s ON d.dept_id=s.dept_id
         GROUP BY d.dept_id,d.dept_name
         HAVING COUNT(s.student_id)=(
           SELECT MAX(cnt) FROM (
             SELECT COUNT(*) AS cnt FROM students
             WHERE dept_id IS NOT NULL GROUP BY dept_id
           ) AS counts_by_department
         );
┌───────────┬───────────────┐
│ dept_name │ student_count │
│  varchar  │     int64     │
├───────────┼───────────────┤
│ ECE       │             3 │
│ CSE       │             3 │
└───────────┴───────────────┘
```



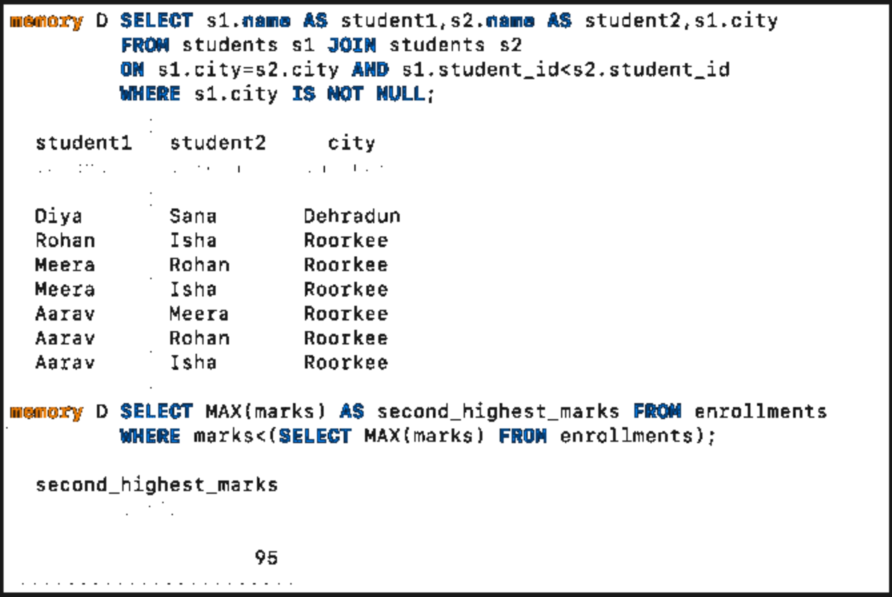
```
memory D SELECT s.name,c.title,e.marks FROM enrollments e
         JOIN students s ON s.student_id=e.student_id
         JOIN courses c ON c.course_id=e.course_id
         WHERE e.marks>(
           SELECT AVG(e2.marks) FROM enrollments e2 WHERE e2.course_id=e.course_id
         );

   name             title          marks


  Aarav     Machine Learning         91
  Kabir     Digital Electronics      86
  Meera     Operating Systems        89
  Sana      Digital Electronics      79
  Meera     Data Structures          95
  Aarav     Data Structures          88

memory D CREATE INDEX idx_students_city ON students(city);
memory D EXPLAIN SELECT * FROM students WHERE city='Roorkee';

┌───────────────────────────┐
│┌─────────────────────────┐│
││       Physical Plan     ││
│└─────────────────────────┘│
└───────────────────────────┘
┌───────────────────────────┐
│         PROJECTION        │
│    ────────────────────   │
│         student_id        │
│            name           │
│            age            │
│            city           │
│          dept_id          │
│                           │
│          ~2 rows          │
└─────────────┬─────────────┘
┌─────────────┴─────────────┐
│          SEQ_SCAN         │
│    ────────────────────   │
│           Table:          │
│    memory.main.students   │
│                           │
│   Type: Sequential Scan   │
│                           │
│        Projections:       │
│            city           │
│         student_id        │
│            name           │
│            age            │
│          dept_id          │
│                           │
│          Filters:         │
│       city='Roorkee'      │
│                           │
│          ~2 rows          │
└───────────────────────────┘
```



```
memory D CREATE VIEW student_marks AS
         SELECT s.name AS student_name,c.title AS course_title,e.marks
         FROM students s JOIN enrollments e ON s.student_id=e.student_id
         JOIN courses c ON e.course_id=c.course_id;
memory D SELECT * FROM student_marks WHERE marks>80;

  student_name      course_title         marks


  Aarav          Data Structures            88
  Kabir          Digital Electronics        86
  Meera          Operating Systems          89
  Aarav          Machine Learning           91
  Meera          Data Structures            95
```



```
memory D START TRANSACTION;
memory D UPDATE enrollments SET marks=100 WHERE student_id=101 AND course_id=201;
memory D ROLLBACK;
memory D SELECT marks FROM enrollments WHERE student_id=101 AND course_id=201;

  marks


     88

memory D START TRANSACTION;
memory D UPDATE enrollments SET marks=100 WHERE student_id=101 AND course_id=201;
memory D COMMIT;
memory D SELECT marks FROM enrollments WHERE student_id=101 AND course_id=201;

  marks


    100

memory D SELECT s.student_id,s.name,SUM(e.marks) AS total_marks,
                RANK() OVER (ORDER BY SUM(e.marks) DESC) AS student_rank
         FROM students s JOIN enrollments e ON s.student_id=e.student_id
         GROUP BY s.student_id,s.name;

  student_id    name     total_marks   student_rank


         101   Aarav             267              1
         104   Meera             184              2
         102   Diya              139              3
         103   Kabir              86              4
         106   Sana               79              5
         108   Isha               71              6
         105   Rohan              58              7

memory D SELECT s.name,c.title,e.marks,
          CASE WHEN e.marks>=90 THEN 'A'
               WHEN e.marks>=75 THEN 'B'
               WHEN e.marks>=60 THEN 'C'
               ELSE 'D' END AS grade
         FROM enrollments e JOIN students s ON s.student_id=e.student_id
         JOIN courses c ON c.course_id=e.course_id;

   name             title            marks    grade


  Aarav     Machine Learning         91    A
  Diya      Machine Learning         72    C
  Kabir     Digital Electronics      86    B
  Meera     Operating Systems        89    B
  Rohan     Thermodynamics           58    D
  Sana      Digital Electronics      79    B
  Isha      Digital Electronics      71    C
  Aarav     Operating Systems        76    B
  Diya      Data Structures          67    C
  Meera     Data Structures          95    A
  Aarav     Data Structures         100    A
```



```
memory D SELECT s1.name AS student1,s2.name AS student2,s1.city
         FROM students s1 JOIN students s2
         ON s1.city=s2.city AND s1.student_id<s2.student_id
         WHERE s1.city IS NOT NULL;

  student1    student2     city


  Diya        Sana        Dehradun
  Rohan       Isha        Roorkee
  Meera       Rohan       Roorkee
  Meera       Isha        Roorkee
  Aarav       Meera       Roorkee
  Aarav       Rohan       Roorkee
  Aarav       Isha        Roorkee

memory D SELECT MAX(marks) AS second_highest_marks FROM enrollments
         WHERE marks<(SELECT MAX(marks) FROM enrollments);

  second_highest_marks


                    95
```